

The economics of charter schools

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Glossary

Charter schools Schools that receive public funding under a contract, or charter, with a public entity (e.g., school district, state, and universities), and are given greater autonomy than other public schools over curriculum, instruction, and operations

External validity An estimate of an effect that is valid beyond the sample of observations included in the analysis

Fixed effects Econometric method of estimating time-invariant factors in panel data analysis

Internal validity An estimate of an effect that is valid only to the sample of observations included in the analysis

Selection bias Estimated effect of programs could be biased because participants have chosen to participate, which is not captured by the econometric model

Introduction

Charter schools are an educational reform in the United States that provide more school alternatives to parents and more local autonomy for individual schools. Individual charter schools receive public funding under a contract, or charter, with a public entity, also known as the charter authorizer (e.g., school district, state, or university). Charter schools are given greater autonomy than traditional public schools over curriculum, instruction, and operations in exchange

for greater accountability for results — charters of charter schools who fail to meet expectations can be non-renewed, effectively closing the school.

The first US charter school opened in Minnesota in 1992. The scale of the charter movement has grown to nearly 7000 schools and three million students in 43 states plus the District of Columbia as of the 2016-17 school year. Charter enrollments constitute about 6% of annual public school enrollments in the United States with enrollment shares as high as 16% and 43% for Arizona and the District of Columbia, respectively.

Charter reform has been a contentious education and public policy issue. Proponents argue that traditional public schools are burdened by excessive centralized control and regulation and greater school autonomy under charter reform allows schools to innovate and improve (Finn, Manno, & Vanourek, 2000). School choice is heralded as a key component of charter success — charters must compete for students with traditional public schools, and their survival is predicated on their ability to implement programs that attract students. Students and parents are afforded more options which creates healthy competition for nearby schools to also improve.

Critics see charters as drawing resources away from traditional public schools and distracting them from internal reforms and innovation. These opponents argue that charter schools are no more effective than traditional public schools, that they create fiscal strains for school districts, that too many of them are unreliably operated, and that they would fail to serve all populations of students, possibly exacerbating racial segregation (Wells et al., 1998). Charter competition is seen as inconsequential in many cases or as counterproductive when traditional public schools placate parents with programs that do little to enhance student learning.

The emergence of charter schools in the United States has paralleled movements in other countries that enacted legislation to create foundation or independent schools that operate much like charter schools in the United States — that is, they receive substantial public funding and have independent authority over curriculum and governance (Brewer & Hentschke, 2008). A key difference between these independent schools and traditional public schools is that enrollment is generally tied to parental choice. As with charter schools, the choice aspect of these schools means that their existence is directly tied to their ability to serve parents and their children.

Policy questions

The extant literature has examined the comprehensive effects of charter schools on the educational system in the United States. Much of the research has narrowly focused on how charter schools affect achievement for students who attend these schools. However, it is important to consider the possible impact charter schools are having more broadly, including their cost effectiveness, systemic effects (e.g., effects on students who choose not to attend traditional public schools), distributional effects, both by ability and race, and any operational differences

between charter and traditional public schools, which may lead to broader educational innovations. This article primarily focuses on four issues:

1. What are the demographic profiles of students entering charter schools? How do charter schools affect the mix of students in traditional public schools by race/ethnicity and academic ability?
2. Are charter schools and traditional public schools receiving comparable funding? Do charters have the same public resources to educate students as traditional public schools?
3. What is the effect of charter schools on the achievement of their students?
4. Do charters promote improved achievement in traditional public schools by introducing competition, or do they harm the performance of traditional public schools?

When summarizing the literature across these questions, we highlight why these questions are important and synthesize the findings across the literature, bearing in mind the quality of the research.

What types of students do charter schools serve?

As charter schools are schools of choice, it is important to examine whether they are serving the full range of the student population. Table 39.1 provides a snapshot of selected characteristics of students served in traditional public schools and charter schools in the 2015–16 school year (USDOE, 2017). On average, charter schools serve more black and Hispanic students than traditional public schools and tend to serve students on further ends of the income distribution than traditional public schools. Over half of the nation's charter schools are located in urban settings as compared to about a quarter of traditional public schools.

TABLE 39.1 Characteristics of traditional public school and charter school students, 2015–16.

	Traditional public schools	Charter schools
<i>Race/Ethnicity</i>		
White	49.9%	33.1%
Black	14.7%	26.8%
Hispanic	25.6%	31.7%
Asian/Pacific Islander	5.4%	4.3%
American Indian/Alaska Native	1.0%	0.7%
Two or more races	3.4%	3.4%
<i>Percent of students eligible for free or reduced-price lunch program</i>		
0%–25.0%	19.6%	22.2%
25.1%–50.0%	27.4%	18.0%
50.1%–75.0%	26.9%	18.6%
75.1%–100%	23.9%	32.6%
<i>Locale</i>		
City	24.9%	56.5%
Suburban	32.1%	25.9%
Town	14.1%	6.7%
Rural	28.9%	10.9%

Source: United States Department of Education, National Center for Education Statistics (2017).

Charter school critics argue that charter success might be illusory if they further stratify an already ethnically or racially stratified system (Cobb & Glass, 1999; Wells et al., 1998). In general, these critics fear that charter schools may not only have negative consequences for the charter students who attend these schools, but may also have social and academic consequences for students in affected traditional public schools. However, proponents of charter schools argue that charter schools will improve racial integration by letting families choose schools outside of neighborhoods where housing may be racially segregated (Finn et al., 2000; Nathan, 1998).

Several studies have examined the racial/ethnic makeup of charter schools relative to the

average racial/ethnic makeup of their surrounding districts and states (Frankenberg & Lee, 2003; Miron & Nelson, 2002; International, 2000; Powell, Blackorby, Marsh, Finnegan, & Anderson, 1997; RPP). While this research has provided some insights, it uses cross-sectional, snapshots of schools' enrollments, which do not permit examination of students moving between schools. Understanding how charter schools affect the mix of students requires a dynamic model that uses longitudinal data to examine the characteristics of students who migrate from a traditional public school to a charter school and compares the students' characteristics with the distribution of students at the old and new schools.

Several studies have examined the migration patterns of students of different race/ethnicity between traditional and charter schools. [Bifulco and Ladd \(2007\)](#) found that black students in North Carolina were likely to switch to charter schools with higher concentrations of black students than the traditional public schools they left. This charter school migration increased the racial isolation of black students. [Booker, Zimmer, and Buddin \(2005\)](#) also found that in California and Texas, black students generally transferred to charter schools with higher concentrations of black students. In contrast, Hispanic students in both states are likely to transfer to charter schools with a lower concentration of Hispanics than in the traditional public schools they left. These patterns have been echoed in a study by [Zimmer et al. \(2009\)](#) across two states and five urban districts across the United States. In Texas, [Weiher and Tedin \(2002\)](#) found that students of white, black, and Hispanic races all transferred into charter schools with greater proportions of their own race than the traditional public schools they left.

Using similar or more rigorous strategies, researchers have also investigated another critique of charter schools – that any success of charter schools is driven by selectively enrolling students through cream skimming high ability students from traditional public schools or pushing out lower-performing students ([Ravitch, 2012](#)). Using data from California and Texas, [Booker et al. \(2005\)](#) compare the prior performance levels of students who move into charter schools with the students at the traditional public schools they leave and find that those moving into charters are generally lower-performing than their previous peers. Similar analyses in a study examining charter schools in Ohio, Texas, and five other major US cities revealed similar conclusions ([Zimmer et al., 2009](#)). [Zimmer and Guarino \(2013\)](#) investigate the claim of pushing out low-performing students in an anonymous major urban school district. They find no evidence consistent with this practice. Two other

studies ([Kho & McEachin, 2017](#); [Kho & Zimmer, 2017](#)) investigate both claims of cream skimming and pushout in Tennessee and North Carolina's charter schools. Neither studies find evidence consistent with such selective enrollment strategies according to student performance level. However, both find some evidence consistent with charter schools' pushout behaviors based on students' disciplinary records. Overall, claims regarding systematic selective enrollment of students, at least according to academic ability, have been unsubstantiated. It is possible that individual charter schools may participate in such practices, but the evidence shows that this would not appear to be a systemwide behavior.

Are charter and traditional schools receiving comparable funding?

Charter schools are nominally entitled to funding comparable with traditional public schools, but some evidence suggests that charter school funding is lagging behind in many states ([Batdorff et al., 2014](#); [Osberg, 2006](#); [Schneider & Tice, 2007](#)). A recent study ([Batdorff et al., 2014](#)) compares the average funding per pupil for traditional public schools and charter schools in 30 states and the District of Columbia and finds that average funding per pupil is about 28.4% lower for charter students, with discrepancies over 40% in some states. In only one state observed is funding comparable across the two sectors – Tennessee. The extent of these funding gaps is largely a function of state policies, including whether the state provides funding for facilities (or requires districts to provide facilities), whether state funds are supplemented by local resources to pay for education, and whether the state paperwork limited charter access to a large array of categorical funds.

In many states, charter schools are expected to pay for facilities out of the operating revenue, which creates large funding gaps ([Krop & Zimmer, 2005](#)). In addition, when states rely heavily

upon local tax revenue to pay for education, charter schools are less likely to gain a share of these revenues — again creating large funding gaps (Osberg, 2006). Finally, some states require education agencies to complete complex forms to receive categorical funding. In school districts, the cost of completing forms is spread over a large enrollment, but autonomous charter schools must complete comparable paperwork to receive funding for small numbers of students. This diseconomy of scale for charters reduces their access to categorical programs compared with traditional public schools and contributes to the reduced funding level of charter schools (Zimmer et al., 2003).

The challenges of these funding gaps are gradually being recognized by states and some new policies have been introduced to address these issues. However, some argue that since one of the objectives of charter schools is greater efficiency, charter schools should prove they can provide similar or better results with fewer resources.

How do charter schools affect the performance of charter students?

Over the last two decades, a series of papers has emerged examining student achievement in charter schools in various states and cities throughout the U.S. In addition, several studies have examined student achievement in charter schools nationally (Carnoy et al., 2005; CREDO, 2009; Cremata et al., 2013; Gleason, Clark, Tuttle, & Dwyer, 2010; Hoxby, 2004; Nelson, Rosenberg, & Van Meter, 2004). Some of this research has relied upon school-level data (Greene, Forster, & Winters, 2003; LOEO, 2003; Miron, Nelson, & Risley, 2002; Rogosa, 2003; Russo, 2005) or cross-sectional comparisons of achievement in charter schools and traditional public schools at a single point in time.

A key weakness of a school-level analysis is the high degree of aggregation, which masks

changes over time in the school's population of students, and variation of performance across different subjects and grades. In essence, school-level data may not pick up the nuances of school characteristics and can only provide an incomplete picture of why outcomes vary across schools. Meanwhile, point-in-time data, even at the student level, cannot account for the amount of time spent in different schools and cannot factor out the various non-school forces at work.

Both school-level and point-in-time studies are likely to provide misleading indications of charter effectiveness due to a selection bias in charter-school enrollment. Parents choose charter alternatives because the traditional schools are not meeting their learning objectives. This parent selection process means that charter students are not representative of the pool of traditional school students, so the average difference in the performance of charter and traditional school students is a misleading indication of the effectiveness of charter schools. For example, if a charter school focuses on low-achieving students that are struggling in traditional schools, then the charter's average test score is likely to reflect the weak preparation of its incoming students. Alternatively, if a high school charter's curriculum is built around college preparation, then the average charter test score may reflect the stronger preparation of its incoming students than that of a nearby traditional high school serving a broader mix of students. The danger is that charter effectiveness measures are easily confounded by the mix of student background and non-school factors that lead parents to choose charter schools for their students.

Reliable measures of charter performance must assess the additional gain or value-add from charter enrollment and not the historical background of students choosing charter schools. Researchers have dealt with the selection bias in charter enrollment in two ways: randomized experiments and longitudinal analyses. Both methods allow researchers to account for

the amount of time a student has spent in a particular school, and both methods address differences among student populations served. Randomized experiments are often considered the gold standard in research, because, by assigning subjects randomly to the treatment condition or control condition, they ensure that differences observed later are the result of treatment rather than the result of background differences between the subject groups.

Several studies examine oversubscribed charter schools that randomly admit students through lotteries. For instance, [Hoxby and Rockoff \(2005\)](#) examined four charter schools in Chicago, which provided evidence that this set of charter schools is outperforming traditional public schools. Later, [Hoxby and Murarka \(2007\)](#) used a similar design to evaluate 47 charter schools in New York City and found a small positive achievement effect for students attending charter schools. More recent studies using lotteries in Boston, Washington DC, and New York City ([Abdulkadiroglu et al., 2011](#); [Curto & Fryer, 2014](#); [Dobbie & Fryer, 2011](#)) have all found positive effects for oversubscribed charter schools.

These studies have strong internal validity — in other words, the results have strong implications for schools included in the evaluation. One drawback of this randomized-design approach is that while it produces valid and reliable results for the set of charter schools examined, it may have limited implications for charter schools that are not part of the analysis. In other words, these studies have weak external validity. In order to conduct a randomized experiment of charter schools, researchers have relied upon randomly assigning students for schools that are oversubscribed with substantial wait lists. Therefore, these studies may have limited implications for those schools that do not have wait lists. In fact, one would expect schools with wait lists to be the best schools and it would be surprising if they had the same results as other charter schools.

When randomized designs have not been possible, or when researchers wanted to be more inclusive in their analysis of charter schools, researchers have often used a student fixed-effect approach to deal with the selection bias, which requires longitudinal student-level data. Student fixed-effect approaches minimize the problem of selection bias by examining the academic gains made by individual students over time, factoring out those students' baseline achievement levels. Moreover, they permit within-student comparisons of achievement gains, examining changes in the achievement trajectories of individual students who move from traditional public schools to charter schools, or vice versa.

To date, a handful of studies have used this approach across a select number of states. In North Carolina, [Bifulco and Ladd \(2006\)](#) used the fixed-effect approach and found negative reading and math effects for its charter schools. These results were supported in later research by [Carruthers \(2012\)](#) after including an additional five years of data. In California, Zimmer and colleagues (2003) examined student-level data from six large districts and found that charter schools were performing on par with traditional public schools. In addition, [Betts, Rice, Zau, Tang, and Koedel \(2006\)](#) examined charter schools exclusively in San Diego and similarly found that charter schools were performing on par with traditional public schools. A cross-state study ([Zimmer et al., 2009](#)) looked at charter schools in two states and five major cities in the US and found a mix of negative effects (in Chicago and Texas) and effects comparable between charter schools and traditional public schools (in Denver, Milwaukee, Philadelphia, San Diego, and Ohio).

Three separate studies using similar approaches found mixed results in Texas. [Gronberg and Jansen \(2001\)](#) found that charter schools that focus on at-risk students provided slightly more value-add than traditional public schools while non-at-risk charters provided

slightly less value-add than traditional schools. Hanushek, Kain, and Rivkin (2002) found negative achievement effects for Texas charter schools that were in the first few years of their operation, however, and no significant effect in reading and small positive effect in math in the long run. In a more recent study, Booker, Gilpatric, Gronberg, and Jansen (2007) found that in the first year of attending charter schools, student test scores dropped, but the students' performance became comparable with students in traditional public schools by the second and third year in reading and math, respectively. In Florida, Sass (2006) found that charter schools initially had lower performance than traditional public schools, but eventually produced similar educational gains over time and by the fifth year, these schools were generally producing small positive effects. Similarly, in Arizona, Solomon, Paark, and Garcia (2001) found that students initially had lower scores, but students who remained 2–3 years in charter schools outperformed traditional public school students in reading and performed at least as well as public school students in math and possibly better (depending upon the model). In Milwaukee and Indianapolis, Witte, Weimer, Shoher, and Schlomer (2007) and Nicotera, Mendiburo, and Berends (2009), respectively, found positive effects for charter schools in math. Finally, in an anonymous district, Imberman (2011) found mixed results for charter schools with a small positive effect in math and a small negative effect in reading.

While the fixed-effect approach can be a strong control for selection bias, there are drawbacks to the fixed-effect approach as well. One of the greatest drawbacks is that the fixed-effect approach provides an estimate of student achievement only for students who switch from a traditional public school to a charter school or vice versa (switchers). Students who remain in either charter or traditional public schools for the duration of the analysis do not contribute to the estimate because the analysis

does not compare their results in both contexts (charter schools and traditional public schools).

Work by Ballou, Teasley, and Zeidner (2007) shows how the reliance on switchers can create two problems. First, charter schools that have high turnover rates (which may be associated with poor performance) will disproportionately contribute to the fixed-effect estimate. Second, Ballou et al. (2007) note that switchers may differ from nonswitchers in important ways, possibly biasing results that rely exclusively on switchers. They explore these issues by examining charter schools in Idaho using both a longitudinal student fixed-effect model, which only uses switchers, and by estimating the impact using all tested charter and noncharter students. Exploring both approaches, they find conflicting results. The authors argue that the bias from only examining switching students in a student fixed-effect model may be greater than the self-selection bias the model attempts to correct and may imply that researchers should not exclusively rely upon fixed-effect models. However, in an interesting counterexample, Betts et al. (2006), using data from San Diego, found that a fixed-effect approach produces math results much closer to the result of a randomized-design approach than other nonexperimental designs. In reading, random-effects models produced the best results.

A third empirical strategy researchers have used recently for charter school evaluation is matching. In this approach, charter school students are matched and compared to students in traditional public schools who look similar on observable characteristics and may have the same propensity to attend a charter school. This approach addresses the problem of student fixed-effect models in that students who have always been enrolled in charter schools or traditional public schools can be included in the estimation, not just switchers. Miron, Cullen, Applegate, and Farrell (2007) use a matching strategy to evaluate charter schools in Delaware

and find negative effects for elementary schools but positive effects for middle and high schools. The Center for Research on Educational Outcomes (CREDO) has released multi-state studies also using a matching strategy (CREDO, 2009; Cremata et al., 2013), finding mixed results that vary by state. In the earlier study, which includes 15 states and the District of Columbia, they find overall negative results. In the latter study, which includes 27 states and the District of Columbia, they find overall positive results.

Several studies have incorporated matching strategies with other methods. Dobbie and Fryer (2016) examine charter schools in Texas using matched samples in combination with fixed effects to evaluate charter schools in Texas. They find that charter schools perform on par with traditional public schools. Berends, Watral, Teasley, and Nicotera (2008) use matching with random effects in a study in an anonymous urban school district and find that charter schools perform worse than traditional public schools in initial years of enrollment, but those effects become positive as the number of years enrolled in the charter school increases.

Matching also has its drawbacks, most commonly discussed is that students are only matched based on observable characteristics. Other characteristics for which data is not available, like motivation, may actually drive students' decisions to enroll in charter schools and be highly correlated with their achievement. By omitting this factor in the matching process, it is unclear whether the matched charter and traditional public school students are truly comparable. Overall, no methodology is perfect. The concerns with each approach suggests that researchers need to be explicit about the types of students the model is examining and what possible bias is remaining when examining charter schools.

Overall, the best of the current research has found that the student achievement results charter schools vary greatly across settings. However, much of this research has shown that

charter schools do improve over time, which implies that we may have to wait longer to draw definitive conclusions about the performance of charter schools.

Is charter school competition improving the performance of traditional public schools?

While much of the existing research on charter school has focused on student achievement effects for students who choose to attend charter schools, we argue that this focus may be too narrow. Supporters hope that charter schools can exert healthy competitive pressure on the existing K-12 educational system by giving families alternatives to traditional public schools. In fact, given that charter schools will probably never educate a substantial portion of the nation's student population, charter advocates argue that these schools may have their greatest impact through systemic effects — the competitive effects of charter schools could improve the performance of traditional public schools and enhance the performance of students who do not attend charter schools.

The challenge in evaluating possible competitive effects is in knowing when district or school personnel will perceive a competitive threat. Do charter schools create competitive pressure when they are located near a traditional public school or when they first appear in a district? Do charter schools only create competitive pressure when they start recruiting students away from a particular school, or do they exert pressure when they capture a certain portion of students within a marketplace? In addition, the local environment may influence the competitive pressure that charter schools create. For instance, some districts may have well-developed, preexisting choice programs, including magnet schools or open enrollment policies. Moreover, some districts may be experiencing significant growth or already have overcrowded schools,

in which case charter schools may act more like a release valve than a source of competitive pressure.

Nevertheless, research is developing around this issue and has made critical assumptions about the competitive process, which could affect conclusions made about the competitive effects. For example, Hoxby (2002) defines competition as where there is minimum market penetration of charter schools within a district and finds substantial positive competitive effects in Arizona and Michigan. However, Bettinger (2005), using an instrumental variable strategy, also examines competitive effects in Michigan as measured by distance and finds no effects. Using school-level data in North Carolina, Holmes, DeSimone, and Rupp (2003) also used distance as a proxy for competition and found substantial competitive effects. In contrast, Bifulco and Ladd (2006) use student-level data in North Carolina and map out the distances of students exiting public schools to enter charter schools. Using this mapping, they analyze the effect charter schools have on traditional public schools within concentric distances of charter schools. Their analysis finds no competitive effects. Sass (2006) and Booker, Gilpatric, Gronberg, and Jansen (2008) also use student-level data in Florida and Texas, respectively, to examine competitive effects. Similar to Bifulco and Ladd, Sass uses concentric circles around public schools and measures whether a charter school is within these concentric circles and what proportion of total students are enrolled in charter schools. Using these approaches, Sass (2006) finds small positive competitive effects in Florida. Booker et al. (2008) uses two approaches, which find consistent competitive effects. First, like Hoxby, the authors use market-penetration measure at the district level. Second, they also use a campus-level market-penetration measure, which is defined by the percentage of students at a particular campus that exits the school to go to a charter school. They find competitive effects across both measures.

In a more recent paper, Buddin and Zimmer (2009) examine the competitive effects of charter schools using both survey and longitudinal student-level data in California. Examining survey responses from traditional public school principals about the changes in school operations, the analysis by the authors suggests that charter schools have no measurable impact on the performance or operation of traditional public schools. The authors go on to examine the student achievement data, in which they use metrics of distance to the nearest charter schools along with the number of charter schools and the share of students attending a charter school within 2.5 miles of a traditional public school, while controlling for the level of preexisting competition and also using various measures of competition. Again, the authors found no evidence that charter schools create a competitive effect. Across these studies, the authors generally found limited evidence of a competitive effect from charter schools. However, that could be explained by the fact that charter schools generally represent a low share of students in districts or states or by the fact that charter schools, at least in growing districts, may act as a release valve rather than a competitive pressure. It is possible that a broader implementation of charters would exert pressure on traditional public schools to improve their performance.

Conclusion

The charter movement grew out of a hope that by providing greater autonomy to schools, charter schools would be able to cut through bureaucratic frustrations and offer innovative, efficient, and effective educational programs, provide new options to families, and promote healthy competition for traditional public schools. However, critics feared that charter schools would drain scarce resources from the existing public education system, create greater stratification both by ability and race/ethnicity,

and provide no real academic advantages to students attending these schools.

The current research has not shown charter schools to be a panacea for challenges facing the educational system that advocates hoped for, but expanding charter numbers and enrollments indicate that these schools may be meeting other academic and/or other nonacademic needs without requiring as many public resources. As the sector continues to grow and charters enter into new facets of the education system, their effects may become more prominent. Researchers should continue to assess and evaluate the charter movement with a greater focus on what particular innovations are and are not driving results for students.

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